

Location-Specific Crash Risk Warnings Using RAG on Historical Crash Data

Progress Update

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- Problem Formulation
- Related Work
- Data
- Proposed Approach
- Experiments Conducted
- Future Work
- Summary

Problem formulation

- UK Roads: <1700 fatal crashes per year
- Drivers have no access to historical, location-specific crash history
- Google Maps
 - relies on real time user reports with no context
- STATS19
 - Dataset containing information but structured for analysts

Problem formulation

● The Gap

- Lacks a connection between historical crash data and driver real time needs for risk analysis

● Existing Systems

- Predict severity as a label, not guidance
- Explainability exists but uses live sensors for AVs

● Hypothesis

- A RAG system using historical crash records can produce location-specific context-aware warnings quickly when needed

● Crash Classification Systems

- Zeng & Yang (2025): tabular + narrative fusion, Macro F1: **0.45** vs CatBoost **0.41**, **0.39** when narratives removed
- Zeng & Yang (2025): fine-tuned LLaMA3-8B, Macro F1: **0.73** vs CatBoost **0.72** with gradient scores

● LLM Explanation of Driving

- Li et al (2025): RAG + vision-based model for AV safety recommendations
- Live + no historical data used
- Location-based retrieval not used

● Overview

- UK national road crash database
- Standardized form (STATS19) recorded by police

● Dataset Content

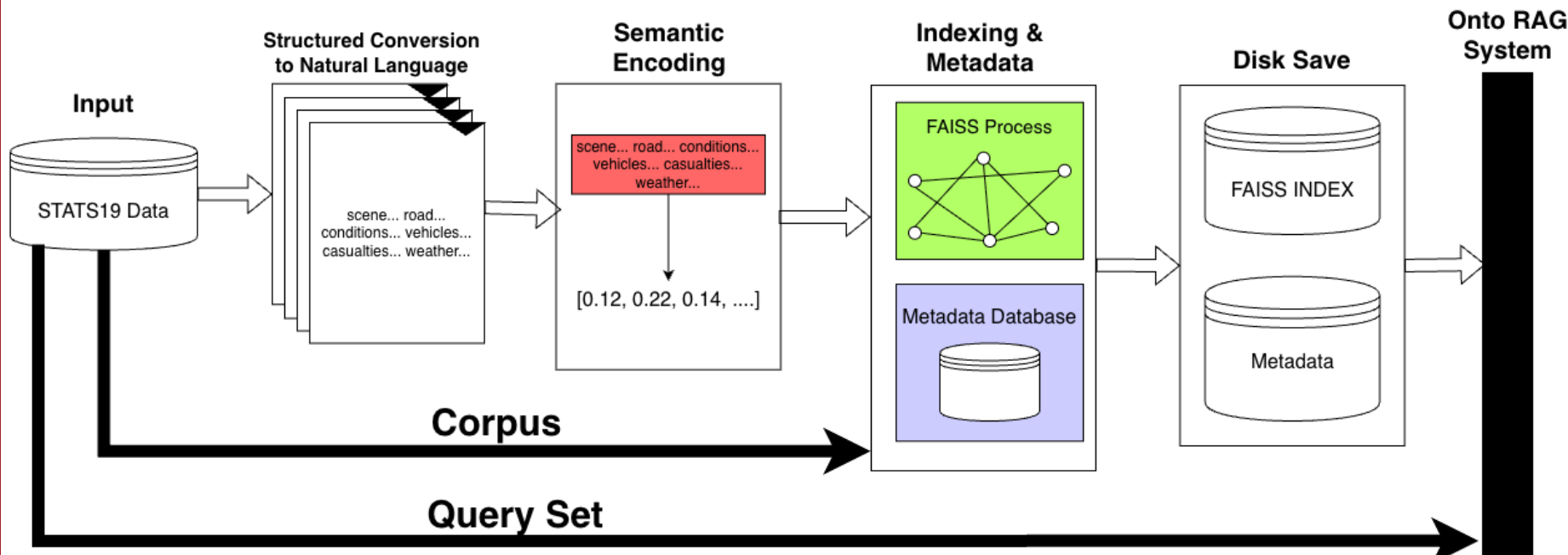
- Collisions: road type, location, weather etc.
- Vehicles: vehicle type/age, maneuver etc.
- Casualties: age, gender, severity etc.

● Statistics

- 503,475 data points (2020-2024)
- **3 class severity**: Slight: 386,007, Serious: 109,977, Fatal: 7,491

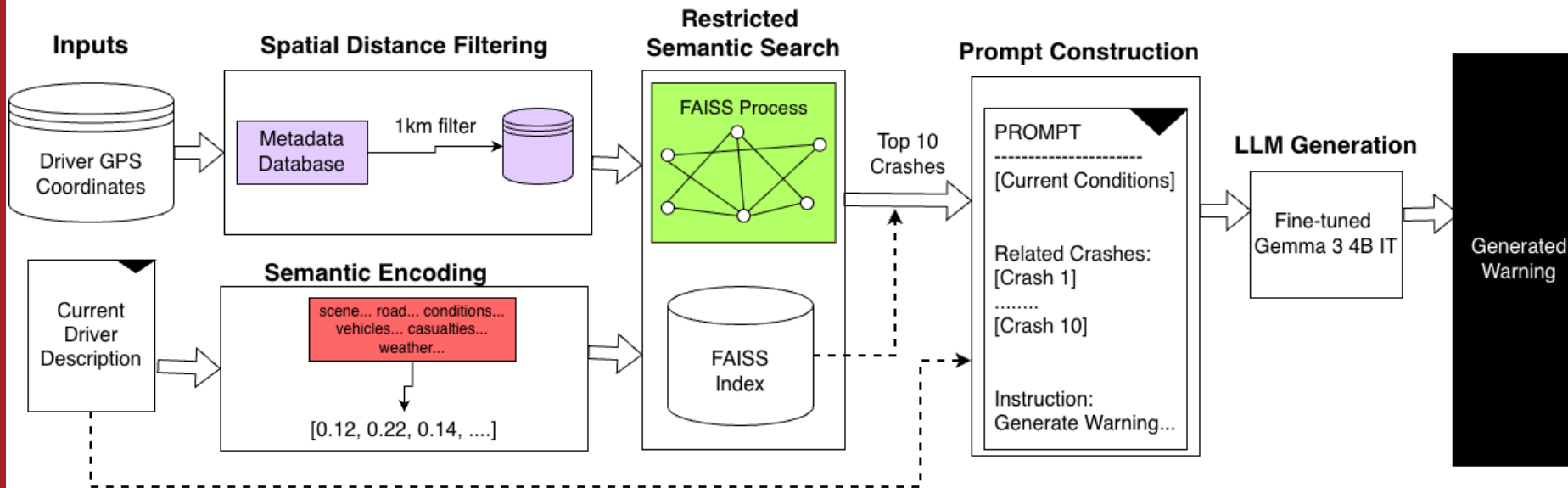
Proposed approach

Preprocessing



Proposed approach

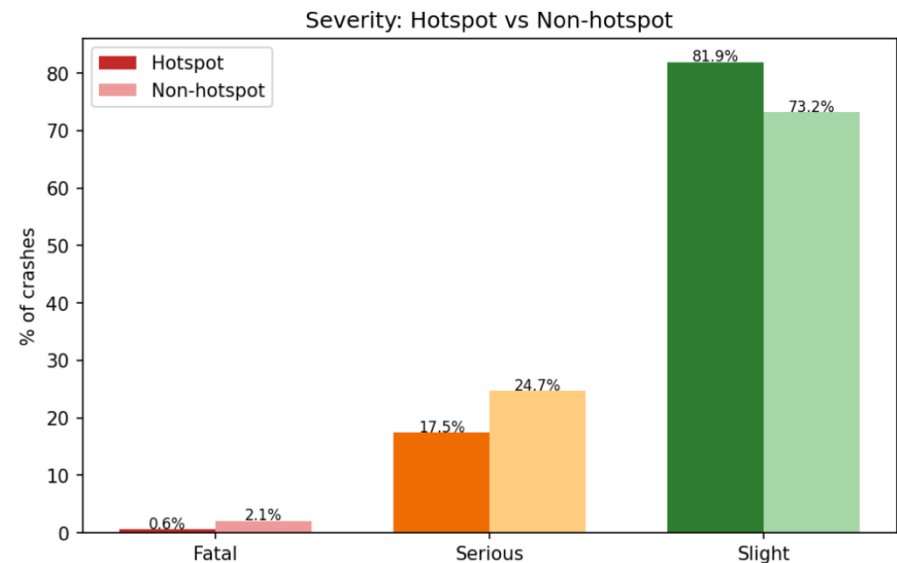
Live System



Experiments Conducted

- Part 1: Fine-tuning
 - Model: Gemma 3 4B IT, QLoRA (r=16)
 - Task: 3-class severity prediction
 - Training set down sampled to lowest class

System	Macro F1	Slight F1	Serious F1	Fatal F1
Zero shot Gemma	0.149	-	-	-
XGBoost	0.350	-	-	-
Fine-tuned Gemma	0.408	0.693	0.364	0.134



Experiments Conducted

- Part 2: RAG System Demo
 - Input: Test Crash Narrative + Lat/Long
 - Spatial Filter -> X crashes within 1km
 - Semantic Search -> top 10 retrieved
 - Fine-tuned Gemma generates location-specific warning

Example Output

Motorcycle collisions are common in urban areas. Always be aware of your surroundings and watch out for motorcyclists, particularly at intersections. Slow down and go slowly.

- Prompt Engineering
 - Force model to surface overrepresented factors
- Evaluation Framework
 - Retrieval quality (MRR, nDCG, Precision/Recall)
 - Warning usefulness (LLM Judge, BERTScore)
- Architecture Tuning
 - Spatial vs no spatial filter
 - Radius (1km / 2km, etc.), top-K (5 / 20, etc.)
 - Sentence transformer variants
- Live Integration & Hotspot Analysis

- Problem

- Drivers do not know location-based harm

- Gap

- Existing work stops at severity prediction or explains AVs only

- Architecture

- End-to-end pipeline -> location-based warnings

- Current Results

- Fine-tuned Gemma has a performance increase

- Future Work

- Evaluation, architecture, prompt engineering